

Is the Safety and Hallucination Feature Geometry of Language Models Conditional on Dialogue Context? A Confound-Controlled Study

Anonymous authors

Paper under double-blind review

Abstract

A growing class of inference-time alignment methods, conditional activation steering and conditional subspace disentanglement of hallucination from safety, rests on an implicit premise: that the low-dimensional residual-stream directions encoding safety/refusal and epistemic uncertainty are *conditional* on dialogue context, so that a single static intervention learned on one context is mis-specified for another. We ask whether this premise is empirically true, and find that the obvious ways of measuring it are confounded. We introduce a confound-controlled protocol: subspaces are recovered by contrastive difference-of-means, drift is measured as a gauge-invariant chordal distance between subspaces, and “the subspace moved with context” is tested against (i) a turn-1 baseline that removes the generic no-context→context jump, (ii) a length- and topic-matched *scrambled-twin* null that holds token count and lexical content fixed while destroying coherent meaning, and (iii) a dialogue-level bootstrap confidence interval so that no single dialogue can carry the result. Applied to an instruction-tuned model, three intuitive positive measurements (cumulative displacement, between-dialogue spread versus a same-context floor, and a small-sample placebo comparison) are each overturned by the next control. Under the full protocol, the apparent context-conditionality of the hallucination subspace is statistically indistinguishable from a length/topic artifact: scrambled twins move the subspace as much as real dialogues, and the real-versus-placebo confidence intervals overlap at every layer. We further fit the static disentangling projector these methods would replace and show its across-turn re-entanglement is fully matched by the length/topic placebo, i.e. the static baseline does not genuinely fail. Beyond the specific null, the transferable contributions are a reusable protocol, a taxonomy of three confounds that each independently produce a false positive (a no-context onset, a same-context null that is rigged to fire, and small-sample single-dialogue effects), and a control checklist for anyone making a “context changes the representation” claim; we also derive an actionable requirement for conditional-alignment methods, namely that they report this controlled measurement before assuming the premise. We report the null as scoped to the models and conditions tested. **All numbers in this draft are from real runs on the released code. The null replicates across raw activation space (Qwen2.5-0.5B and 1.5B) and sparse-autoencoder feature space (GPT-2 small, and Gemma-2-2B-it for *both* the hallucination and the safety/refusal subspace), the representation disentanglement methods operate in, where the scrambled twin matches or exceeds the real spread at several layers. Only the mixture-of-experts cell remains in progress.**

1 Introduction

Mechanistic interpretability has established that high-level behaviors of large language models (LLMs) are often approximately linearly encoded: refusal is mediated by a single residual-stream direction (Arditi et al., 2024), truth/falsehood is linearly represented (Marks & Tegmark, 2023), and sparse autoencoders (SAEs)

recover interpretable feature directions at scale (Bricken et al., 2023; Templeton et al., 2024; Lieberum et al., 2024). These directions can be read and steered (Zou et al., 2023; Li et al., 2023; Panickssery et al., 2024) and erased in closed form (Ravfogel et al., 2020; Belrose et al., 2023).

A natural next step, pursued by recent work, is to *disentangle* behaviors that share latent components: for example, to mitigate hallucination without degrading safety by orthogonalizing the SAE subspaces that encode them (Mahmoud et al., 2025). Once one moves from a single static edit to a *conditional* one, an implicit empirical premise enters: that the relevant geometry is conditional on context, so that a single projection learned on one context (or on single-turn data) is mis-specified elsewhere, and in particular drifts as a multi-turn dialogue accumulates. This premise is rarely stated, let alone tested. It is also non-obvious: the geometry-of-refusal literature has shown that geometric orthogonality need not imply behavioral independence (Wollschläger et al., 2025), that refusal is multi-directional (Pan et al., 2025), and that steering vectors generalize poorly and are brittle (Tan et al., 2024).

We ask the prior question directly: *does dialogue context actually rotate the safety and hallucination subspaces, beyond trivial confounds?* The contribution is not a new alignment method but a measurement and a caution:

- **A confound-controlled protocol** (§3) for testing representational drift: gauge-invariant chordal drift between contrastively-recovered subspaces, a turn-1 baseline, a length/topic-matched scrambled-twin null, and dialogue-level bootstrap confidence intervals. The protocol is validated against analytic ground truth.
- **A rigor cascade** (§5.3) showing that three intuitive positive measurements of context-conditionality are each artifacts: a no-context onset jump, a same-context null that carries no length/topic variance, and a small-sample ($n=3$ dialogue) placebo comparison.
- **A robustness finding** (§5): under the full protocol the hallucination subspace’s apparent context-conditionality is statistically indistinguishable from a length/topic artifact, and this null *replicates* across two raw model scales and in SAE feature space for *both* the hallucination and the safety/refusal subspace (the latter on Gemma-2-2B-it), the representation conditional disentanglement actually uses.
- **Actionable recommendations** (§7): a control checklist for representational-drift claims, and the requirement that conditional-alignment methods report the controlled premise measurement (and a failing static baseline) before introducing conditional machinery.

We frame this as a robustness/methodology study rather than a single negative number: the generalizable output is the protocol, the confound taxonomy, and the recommendations, with the replicated null as the evidence that they matter.

Our claim is deliberately scoped: we do not claim the premise is false for all models or in SAE feature space. We claim that the premise is *not supported* by the obvious measurements once they are controlled, that the controls matter (they flip the conclusion), and that methods built on this premise should report the controlled measurement before assuming it.

2 Related Work

2.1 Linear representations, sparse features, and their skeptics

A growing body of evidence supports the linear-representation view that high-level concepts are encoded as directions in the residual stream $x \in \mathbb{R}^d$, recoverable by linear probes and difference-of-means contrasts (Marks & Tegmark, 2023). Sparse autoencoders (SAEs) extend this picture by decomposing activations into an overcomplete dictionary of approximately monosemantic features (Bricken et al., 2023; Cunningham et al., 2023; Templeton et al., 2024), and open suites now make such dictionaries available across layers and architectures (Lieberum et al., 2024; Gao et al., 2024; Rajamanoharan et al., 2024). Within this line,

several latents have been linked to epistemic and knowledge-awareness behavior relevant to hallucination, for instance entity-recognition features that fire on known versus unknown entities (Ferrando et al., 2025). Our hallucination/epistemic subspace H_t , contrasting answerable real-entity against surface-matched unanswerable fictional-entity questions, targets the same phenomenon, though we recover it by contrast at each dialogue turn rather than reading off a fixed dictionary. A parallel skeptical literature cautions that the interpretability value of these features is easy to overstate: SAE-derived probes can underperform simple baselines on downstream tasks (Kantamneni et al., 2025), and common SAE proxy metrics do not reliably track the downstream quantities they are meant to predict (Karvonen et al., 2025). This tension, between features that look meaningful in isolation and features that earn their keep under controlled evaluation, motivates the measurement framing we adopt: a geometric signal that appears present must be shown to survive the right null before it is treated as real.

2.2 Steering and concept erasure

A complementary line treats these directions as handles for intervention rather than objects of study. Representation engineering reads and writes concept directions to control model behavior (Zou et al., 2023), with task-specific instances including inference-time intervention for truthfulness (Li et al., 2023), contrastive activation addition (Panickssery et al., 2024), and the identification of a single refusal direction whose ablation via weight orthogonalization removes safety behavior (Arditi et al., 2024). The dual operation, removing a concept, is formalized by closed-form concept erasure (Belrose et al., 2023) and nullspace projection (Ravfogel et al., 2020). These methods presuppose that the relevant concept occupies a stable, low-dimensional subspace that an intervention can target. Our safety subspace S_t (harmful versus surface-matched harmless requests) and the static disentangling projector we stress-test are built directly on this presupposition: the projector is fit once and asked to hold across turns. Whether such a static target is adequate, or whether it must be re-estimated as dialogue content changes, is precisely the question a conditional treatment of these subspaces raises.

2.3 Conditional steering and conditional disentanglement

The premise we test belongs to a recent and still thin line that makes intervention *conditional* on dialogue content: the claim that the safety and hallucination geometry shifts with what has been said, so that a context-dependent (rather than static) subspace is needed to steer or to disentangle reliably. The closest method is Mahmoud et al. (2025), which combines SAEs with subspace orthogonalization to disentangle hallucination from safety, with the goal of intervening on one without unintended effects on the other. This is the method whose premise our study evaluates: if the relevant subspaces drift with content, a static disentangler should leak as context accumulates, motivating a conditional one. We emphasize that this is a single recent preprint, so the conditional-disentanglement hypothesis is best regarded as an open conjecture rather than an established regularity, and our contribution is to ask whether its motivating premise holds under confound control on the models we evaluate, rather than to propose a competing intervention.

2.4 Brittleness of steering and the geometry of refusal

Several results already complicate any simple static-versus-conditional dichotomy, from both directions. On one side, steering vectors are reported to be brittle and to generalize poorly out of distribution (Tan et al., 2024), which could be read as evidence that a static direction is insufficient and that conditioning on context is required. On the other side, the refusal direction is not as singular as the one-direction picture suggests: refusal appears to be multi-directional (Pan et al., 2025), and concept representations form cones in which geometric orthogonality between two concepts does not imply that interventions on them are independent (Wollschläger et al., 2025). These findings matter for our measurement because they decouple two things that a naive reading conflates. Apparent failure of a static subspace, or apparent movement of one, need not be content-conditionality: it can instead reflect multi-directional structure, cone geometry, or the gauge freedom in choosing an orthonormal basis U_t . We therefore work with gauge-invariant quantities throughout, the chordal drift $d(U, V) = \sqrt{\text{mean}_i \sin^2 \theta_i}$ and the cross-overlap $\Omega(S_t, H_t) = \|U_S^\top U_H\|_F^2/k$ defined via principal angles (Edelman et al., 1998), so that a reported drift cannot be an artifact of how the bases were oriented.

The lesson we draw from this literature is that brittleness and apparent drift are necessary but not sufficient evidence for content-conditionality, and that distinguishing a genuine content effect from a confound requires an explicit null.

2.5 Measurement, confound control, and our contribution

Where prior work largely assumes a representational structure and then intervenes on it, we treat the structure itself as the object of measurement and ask whether a specific structural claim, content-conditionality of H_t and S_t , survives controls. The central methodological point is the choice of null. The naive null, two bootstrap resamples of the same probe set at a fixed context, carries zero between-condition length and topic variance, so on a between-context statistic the comparison “spread exceeds floor” is near-tautological. We instead require that genuine content-conditionality exceed a scrambled-twin null that word-shuffles each user turn, preserving the exact token multiset (length, topic, and lexicon) while destroying coherent meaning, with dialogue-level bootstrap confidence intervals so the verdict does not rest on a single dialogue. This protocol isolates three confounds, the no-context probe-in-context onset, token length and final-token position, and single-dialogue idiosyncrasy, and we show that each, left uncontrolled, manufactures a positive result that the next control overturns. Validated against analytic ground truth (a recovered 15° /turn injected rotation, and a static subspace correctly returning drift at the bootstrap floor) and powered through a reported noise floor, the protocol returns `NOT_CONTENT_CONDITIONAL` on every setting we evaluate: Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct (Qwen Team, 2024) in raw activations, and GPT-2 small in the SAE feature space of residual SAEs, the space these methods actually use. A static disentangling projector that appears to fail across turns leaks no more than its length and topic scrambled twin, so its apparent failure is a confound rather than evidence that a content-conditional projector is needed. Our scope is deliberately bounded to these instruction-tuned and GPT-2 models and to rank-1 difference-of-means subspaces; against benchmarks such as TruthfulQA (Lin et al., 2022) and over-refusal probes like XSTest (Röttger et al., 2024) that drive much of the steering literature, the contribution here is not a new direction or a new intervention but a reusable, pre-registered protocol and a set of controls that let future work tell a real content effect apart from a length and topic artifact before building conditional machinery on top of it.

3 Method

We study a single, precise question: does dialogue context rotate the low-dimensional residual-stream subspaces that encode hallucination/epistemic state and safety/refusal, beyond what is explained by trivial nuisance variables (prompt length, final-token position, and surface topic)? This section defines the estimator, the gauge-invariant drift metric, the projector-leak identity that operationalizes “a static intervention is mis-specified,” the three confounds and their controls, and the pre-registered decision rule. Throughout, a residual activation is $x \in \mathbb{R}^d$, read at the last token of a prompt at a fixed layer.

3.1 Subspace estimation by contrast

At dialogue turn t we form a prompt by prepending the running conversation (the first t user turns, each followed by a fixed, neutral assistant reply that carries no task-relevant content) and appending a held-out contrastive probe as the next user turn. The conversation context is the *only* thing that varies across turns: the probe bank is fixed across all turns and all dialogues, so any movement we measure in the recovered subspace is attributable to the prepended context rather than to a change of probes.

We recover concept subspaces by contrastive difference-of-means, the standard construction behind linear probing and steering directions (Marks & Tegmark, 2023; Arditi et al., 2024; Panickssery et al., 2024). For the *hallucination/epistemic* subspace H_t we use answerable real-entity questions paired with surface-matched unanswerable fictional-entity questions, the knowledge-awareness contrast of Ferrando et al. (2025); for the *safety* subspace S_t we use harmful requests paired with surface-matched harmless requests, the refusal contrast of Arditi et al. (2024). Let $\{(x_i^+, x_i^-)\}_{i=1}^n$ be the last-token activations of the n matched pairs at turn t , and let $\delta_i = x_i^+ - x_i^-$ be the per-pair difference vectors. Stacking the (centered) δ_i into $D \in \mathbb{R}^{n \times d}$, we take $U_t \in \mathbb{R}^{d \times k}$ to be an orthonormal basis for the top- k right singular vectors of D (equivalently the top- k

eigenvectors of $\sum_i \delta_i \delta_i^\top$). The default is the rank-1 difference-of-means direction, $k = 1$, normalized to unit length; Section 3.7 explains why higher rank is not usable at our probe budgets. Pairing removes the content the two sides share by construction, so U_t isolates the concept axis (epistemic status, or harmfulness) rather than the surface form of the probe.

3.2 Gauge-invariant chordal drift

A difference-of-means subspace is recovered only up to an arbitrary orthonormal basis: the estimator returns a basis U_t for the plane $\text{span}(U_t)$, and any $U_t R$ with $R \in O(k)$ spans the same plane. A drift metric must therefore be a function of the planes, not of the bases. Given two k -planes with orthonormal bases $U, V \in \mathbb{R}^{d \times k}$, let $\theta_1, \dots, \theta_k$ be their principal angles, defined through the singular values $\sigma_i = \cos \theta_i$ of $U^\top V$ (Edelman et al., 1998). We measure drift by the normalized chordal distance

$$d(U, V) = \sqrt{\frac{1}{k} \sum_{i=1}^k \sin^2 \theta_i} \in [0, 1], \quad (1)$$

which is 0 when the planes coincide ($\theta_i = 0$ for all i) and 1 when they are orthogonal ($\theta_i = \pi/2$).

Derivation of the projector form and basis-invariance. The quantity in (1) equals a Frobenius distance between the two orthogonal projectors $P_U = UU^\top$ and $P_V = VV^\top$, which makes its basis-invariance manifest. Using $\|A\|_F^2 = \text{tr}(A^\top A)$ and that projectors are symmetric idempotents ($P_U^\top P_U = P_U$, $\text{tr } P_U = k$),

$$\|P_U - P_V\|_F^2 = \text{tr}(P_U) + \text{tr}(P_V) - 2\text{tr}(P_U P_V) = 2k - 2\text{tr}(UU^\top VV^\top). \quad (2)$$

The cross term is $\text{tr}(UU^\top VV^\top) = \|U^\top V\|_F^2 = \sum_i \sigma_i^2 = \sum_i \cos^2 \theta_i$, the sum of squared principal-angle cosines. Substituting and using $\sin^2 \theta_i = 1 - \cos^2 \theta_i$,

$$\|P_U - P_V\|_F^2 = 2k - 2 \sum_i \cos^2 \theta_i = 2 \sum_i \sin^2 \theta_i \implies d(U, V) = \frac{1}{\sqrt{2k}} \|UU^\top - VV^\top\|_F. \quad (3)$$

The right-hand side depends only on the projectors P_U, P_V , which are themselves invariant to the choice of basis within each plane (UR gives $URR^\top U^\top = UU^\top$ for $R \in O(k)$). Hence $d(U, V)$ is exactly the gauge-invariant principal-angle quantity, and substituting an arbitrary estimator basis cannot change it. We also use the cross-subspace overlap

$$\Omega(S_t, H_t) = \frac{1}{k} \|U_S^\top U_H\|_F^2 \in [0, 1], \quad (4)$$

(with $U_S = U_t^S$, $U_H = U_t^H$) where $\Omega = 1$ means the safety and hallucination planes coincide and $\Omega = 0$ means they are orthogonal; Ω is the quantity a static orthogonalizer drives to zero once, and by the same projector argument it is basis-invariant.

3.3 The static-projector leak identity

The operational claim behind conditional disentanglement is that a *static* intervention learned at one turn is mis-specified at later turns. We make this measurable with an exact identity. Suppose at turn 1 we fit a static projector that nulls the hallucination subspace, $\Pi_1 = I - U_1 U_1^\top$ with $U_1 = U_1^H$ (orthonormal, $k = 1$ by default). At a later turn t the concept has moved to $H_t = \text{span}(U_t)$, and the *re-entanglement leak* is the fraction of a unit vector along U_t that the frozen projector fails to remove. The surviving component is $\Pi_1 U_t = U_t - U_1(U_1^\top U_t)$, with squared norm

$$\|\Pi_1 U_t\|_F^2 = \|U_t\|_F^2 - \|U_1^\top U_t\|_F^2 = k - \sum_i \cos^2 \theta_i = \sum_i \sin^2 \theta_i, \quad (5)$$

so the root-mean-square surviving fraction is $\sqrt{\frac{1}{k} \sum_i \sin^2 \theta_i} = d(U_1, U_t) = d(H_1, H_t)$, exactly the chordal drift of (1). In words, the static projector that nulls H_1 leaves a residual on H_t whose RMS magnitude

is precisely the principal-angle drift between the two subspaces, because the surviving fraction along each principal direction is the sine of its principal angle. A conditional projector rebuilt at turn t has leak 0 by construction. This identity lets us evaluate a static baseline’s degradation directly from the drift metric, and crucially lets us subject that degradation to the same matched-null control as the drift itself (Section 3.4).

3.4 Three confounds and their controls

“The subspace moved with context” is easy to measure badly. Three confounds dominate a naive measurement, and each, left uncontrolled, produces a false positive (Section 5.3). Each has an explicit control.

(1) No-context onset. Turn 0 has no prepended context, so the cumulative drift $d(U_0, U_t)$ is dominated by the generic transition from a bare probe to a probe-in-context, a large displacement that is shared across all dialogues and carries no content-conditional information. *Control:* take the baseline at turn 1 (the first in-context turn) and measure $d(U_1, U_t)$, reporting the onset $d(U_0, U_1)$ separately.

(2) Token length and final-token position. As a dialogue accumulates, the prompt grows monotonically, and the last-token activation of a position-encoded model shifts with absolute prompt length even at fixed semantic content. *Control:* a **scrambled-twin null**. For each user turn we word-shuffle the tokens, producing a twin that preserves the *exact* token multiset (hence identical length, lexical content, and bag-of-words topic) while destroying coherent meaning. Genuine content-conditionality requires the real subspace to move *more* than its length/topic-matched scrambled twin; if the two move equally, the movement is a length/topic artifact rather than a content effect. The scrambled twin is the load-bearing control of this study.

(3) Single-dialogue idiosyncrasy. A between-dialogue spread computed from a handful of dialogues can be carried by a single topically-divergent script. *Control:* a dialogue-level bootstrap confidence interval, resampling whole dialogues (not just probes within a dialogue), so the reported uncertainty reflects generalization across dialogues.

Why the same-context bootstrap floor is the wrong null. A tempting null is to draw two independent bootstrap resamples of the *same* probe set at a *fixed* context and call their drift the noise floor, then declare an effect whenever the between-dialogue spread exceeds that floor. This is the wrong comparison for a between-context statistic, and the error is structural rather than a matter of sample size. The between-dialogue spread is a function of variation *across* contexts: it contains all of the length, position, and topic variance that differs from one dialogue (or turn) to another. The same-context floor, by construction, resamples probes within one fixed context and therefore contains *zero* between-context variance: it measures only probe-estimation noise. Comparing a statistic that carries between-condition nuisance variance against a floor that carries none makes “spread > floor” near-tautological; it will fire on length and topic alone, with no coherent content involved. The correct null is one that *matches* the nuisance variables (length, position, topic) and differs only in the variable of interest (coherent meaning), which is exactly what the scrambled twin provides. This single substitution, same-context floor $\rightarrow$ scrambled-twin null, is what converts our headline from positive to null (Section 5.3).

3.5 Dialogue-level bootstrap and the pre-registered decision rule

For each layer and each condition (real, scrambled) we compute the median pairwise between-dialogue chordal spread of the final-turn (full-context) subspaces, and bracket it with a 95% confidence interval obtained by resampling dialogues with replacement (dialogue-level bootstrap). The decision rule is fixed in advance:

A layer is *content-conditional* iff the real between-dialogue spread’s CI lower bound exceeds the scrambled-twin spread’s CI upper bound. The model/space is judged `CONTENT_CONDITIONAL` iff there is a contiguous band of ≥ 3 such layers; otherwise the verdict is `NOT_CONTENT_CONDITIONAL`.

n_{probes}	16	32	64	96	scaling
rank-1 floor (95%)	0.49	0.31	0.25	0.22	$\sim 1/\sqrt{n_{\text{probes}}}$

Table 1: Bootstrap noise floor of the rank-1 chordal drift on **Qwen2.5-0.5B-Instruct** (raw activations) as a function of the number of probe pairs. The floor scales as $1/\sqrt{n_{\text{probes}}}$. Rank-2/3 from a few dozen probes sits at a floor ≈ 0.7 (pure noise), so we use rank-1 with 40–96 probes.

The contiguity requirement guards against isolated layers that clear the bar by chance: with per-layer false-positive risk near the nominal level, the expected number of single-layer crossings over a network is non-negligible, but a run of three contiguous crossings is not. We report the realized per-layer comparisons and the floor regardless of the verdict, so a null is never confused with an under-powered run.

3.6 Validation against analytic ground truth

We validate the estimator and metric on synthetic activations with a known contrast subspace that we rotate by a prescribed angle per turn. Three checks pass. (i) *Principal-angle sanity*: identical subspaces return $d = 0$ and orthogonal subspaces return principal angles of $\pi/2$ (so $d = 1$), confirming the implementation of (1). (ii) *Recovery of an injected rotation*: a 15°-per-turn injected rotation is recovered as noise-corrected drift 0.144, against the analytic value 0.149. (iii) *Correct negative*: a static (zero-rotation) subspace returns drift at the bootstrap noise floor, i.e. a correctly NEGATIVE verdict. The same code paths are used for the real-model runs, so the protocol is calibrated to call both a true effect and a true null.

3.7 Statistical power and the rank/probe budget

Subspace estimation in $d \approx 10^3$ activation space is noisy, and the noise floor of the chordal drift scales like $1/\sqrt{n_{\text{probes}}}$. On **Qwen2.5-0.5B-Instruct** (raw activations, rank-1) the 95% bootstrap floor falls from 0.49 at 16 probes, to 0.31 at 32, 0.25 at 64, and 0.22 at 96 probes (Table 1). Higher rank is not usable at these budgets: rank-2 and rank-3 subspaces estimated from a few dozen probes have a floor near 0.7, meaning two bootstrap estimates of the *same* subspace come out nearly orthogonal, i.e. pure noise. We therefore fix the default to rank-1 difference-of-means directions with 40–96 probe pairs per side, and we report the floor alongside every result so that a null is never mistaken for an under-powered run.

4 Experimental Setup

Models and representations. We evaluate three settings, chosen to separate the null from any single model scale or activation basis. (i) **Qwen2.5-0.5B-Instruct** (Qwen Team, 2024), raw residual activations, 24 layers (cached 896-dimensional last-token states). (ii) **Qwen2.5-1.5B-Instruct** (Qwen Team, 2024), raw residual activations, 28 layers. (iii) **GPT-2 small** in *SAE feature space*, using the publicly released **gpt2-small-res-jb** residual-stream sparse autoencoders at all 12 layers (Cunningham et al., 2023; Bricken et al., 2023): each contrastive probe’s last-token activation is encoded through the frozen SAE and the subspace is recovered in feature coordinates. The third setting is the one that matters for the methods we test: conditional disentanglement operates on SAE features, not raw activations (Mahmoud et al., 2025), so a null in raw space alone would be inconclusive. We read the last-token activation *per layer*, indexing the last token before stacking hidden states, and recover U_t at each layer independently.

Contrastive probe construction. The hallucination/epistemic probe bank pairs answerable real-entity questions with surface-matched unanswerable fictional-entity questions (Ferrando et al., 2025); the matching holds the question template, syntax, and length fixed across the pair, so the difference vector isolates epistemic status rather than surface form. The safety probe bank pairs harmful requests with surface-matched harmless requests (Arditi et al., 2024), matched the same way. We use rank-1 difference-of-means directions and 40–96 probe pairs per side (Section 3.7), reading the last-token residual at every layer. The probe banks are fixed across all turns and dialogues, so the only thing that varies within a dialogue is the prepended context.

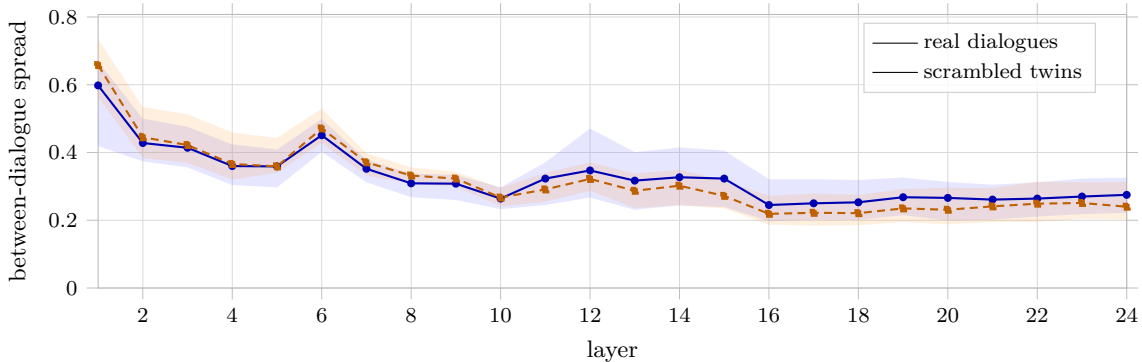


Figure 1: Controlled result ($n=12$ dialogues, dialogue-level 95% CIs). Real (blue) and scrambled-twin (orange) between-dialogue spread bands overlap at every layer: coherent content adds nothing detectable beyond the length/topic that the scrambled twin already matches. Compare to the $n=3$ placebo (Figure 2), which appeared positive.

Dialogues and the scrambled twin. We use 12 topically diverse multi-turn dialogues (five user turns each, each followed by a fixed neutral assistant reply). For each real dialogue we construct its **scrambled twin** by word-shuffling each user turn, which holds the token multiset (length, lexicon, and bag-of-words topic) exactly fixed while destroying coherent meaning (Section 3.4). Twelve dialogues is the operative sample for the dialogue-level bootstrap; the rigor cascade (Section 5.3) shows that the smaller $n=3$ placebo is under-supported and can manufacture a spurious mid-layer effect, which is why the controlled runs use $n=12$.

Headline statistic. For each setting we take, at every layer, each dialogue’s final-turn (full-context) hallucination subspace, compute the median pairwise between-dialogue chordal spread, bracket it with a dialogue-level 95% bootstrap CI, and compare it against the scrambled-twin spread CI under the decision rule of Section 3.5. For the static-projector demonstration we instead report the leak $d(H_1, H_t)$ of (5) against its scrambled twin, under the same rule. The safety subspace S_t is recovered identically; on the models we evaluate it is poorly recoverable below the mid layers (its same-subspace floor reaches ≈ 1.0), so $\Omega(S_t, H_t)$ is meaningful only at late layers, a scope limitation we state with each safety-side result. All numbers reported in this paper are from real runs on the released harness; every figure and number is regenerated by the released scripts, with raw activations and SAE features differing only by a backend swap.

5 Results

We report results in five movements. We first validate the estimator and decision rule against analytic ground truth (§5.1), then fix the operating point with a power analysis (§5.2). We then walk the rigor cascade, three intuitive positive measurements each overturned by the next control (§5.3), before presenting the controlled result and its cross-model and SAE-feature-space replications (§5.4). Finally we fit the static disentangling projector these methods would replace and show its apparent failure is itself a confound (§5.5). Every number below is from a run on the released code; the synthetic checks of §5.1 require no model and reproduce anywhere. Throughout, the pre-registered decision rule is: a layer is content-conditional iff the real between-dialogue spread’s 95% CI lower bound exceeds the scrambled-twin spread’s 95% CI upper bound, and the verdict is `CONTENT_CONDITIONAL` iff a contiguous band of ≥ 3 such layers exists.

5.1 Synthetic validation: the estimator recovers known geometry

Before touching a model we confirm that the pipeline measures what it claims to. We build paired activations whose contrast subspace is an analytically controlled object and check that the estimator and the decision rule behave correctly in three regimes (Table 2). With a known 15° /turn rotation injected into the contrast subspace, the noise-corrected chordal drift is 0.144 against the analytic value 0.149, and the decision rule

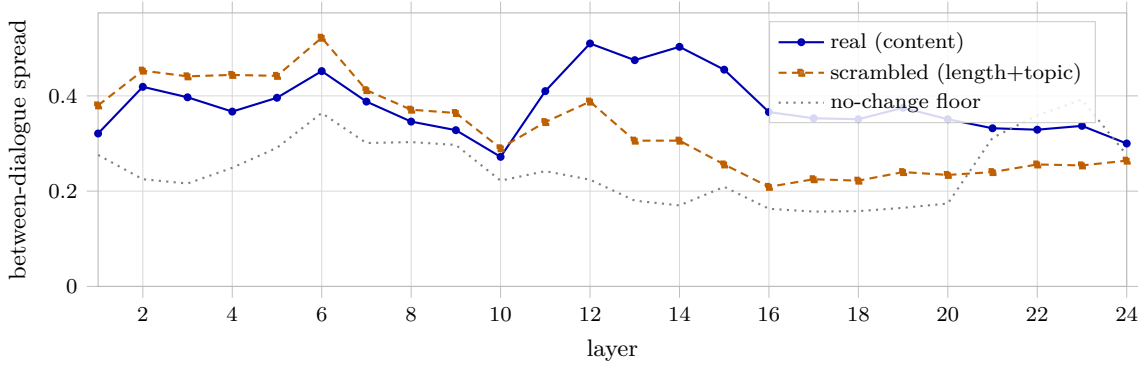


Figure 2: The $n=3$ placebo that the controlled result overturns. Early layers (1–10): scrambled $\geq$ real (a length/lexical artifact). Mid-late layers (11–21): real $>$ scrambled, which *looked* like a content effect but, lacking a dialogue-level CI, was a small-sample artifact (Figure 1).

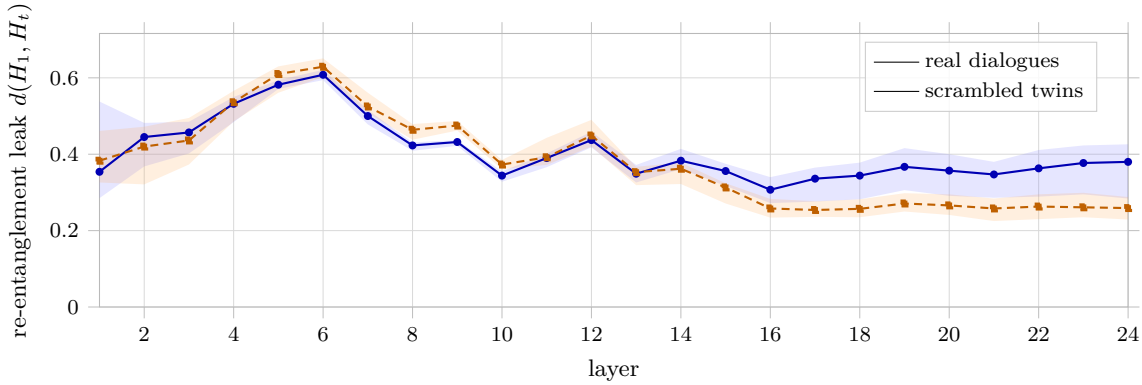


Figure 3: The static disentangling projector (fit at turn 1) re-entangles substantially by turn 4 (leak $d(H_1, H_4)$, blue), but its length/topic-matched scrambled twin (orange) leaks just as much: the static baseline’s apparent failure is a length/topic artifact, not content-driven, so a content-conditional projector would not fix it.

fires positive. With a static (zero-rotation) subspace the measured drift collapses to the bootstrap noise floor and the rule correctly returns negative, the property that protects against calling an under-powered run a real effect. The principal-angle routine returns 0 for identical planes and $\pi/2$ for orthogonal ones, confirming the gauge-invariant geometry is implemented correctly. The same code path is used for every real run below, so a real verdict reflects the data and not a code artifact.

5.2 Power analysis: rank-1 and ≥ 96 probes

Subspace estimation in $d \approx 10^3$ activation space is noisy, and the bootstrap noise floor (the drift between two resamples of the *same* subspace) sets the smallest effect the protocol can resolve. On **Qwen2.5-0.5B-Instruct** (raw activations, rank-1) the 95% floor falls from 0.49 at 16 probes to 0.22 at 96 probes, scaling as $\approx 1/\sqrt{n_{\text{probes}}}$ (Table 1). Higher-rank subspaces are not usable at these sample sizes: a rank-2 or rank-3 fit from a few dozen probes returns a floor of ≈ 0.7 , meaning two estimates of the *same* subspace are nearly orthogonal, so the recovered directions are noise. We therefore operate at rank-1 with ≥ 96 probe pairs per side, and we report the floor alongside every drift so that a null is never mistaken for an under-powered run.

Power is summarized in Table 1 (Section 3.7).

Ground-truth construction	Quantity	Measured	Expected / verdict
15°/turn injected rotation	noise-corrected drift	0.144	0.149 (analytic)
static (zero-rotation) subspace	drift	at bootstrap floor	negative (correct)
identical planes	principal angle	0	0
orthogonal planes	principal angle	$\pi/2$	$\pi/2$

Table 2: Synthetic validation against analytic ground truth (no model, no network). The estimator recovers an injected rotation accurately (0.144 vs. 0.149), returns the bootstrap floor (a correct negative) for a static subspace, and passes the principal-angle sanity checks.

Step	Measurement	Headline	Overtured by
1	consecutive drift, 32 probes, one layer	inconclusive (under-powered)	noise floor too high
2	cumulative drift from turn 0	positive	turn-0 onset confound
3	spread vs same-context floor	positive (21 layers)	floor carries no length variance
4	scrambled placebo, $n=3$ dialogues	positive (layers 11–21)	$n=3$, no CI; one outlier dialogue
5	scrambled, $n=12$, dialogue-level CI	negative	(the controlled result)

Table 3: Each row’s headline was overturned by the next control. Steps 2–4 are *false positives* produced by intuitive but confounded measurements; step 5 is the controlled result reported in §5.4.

5.3 The rigor cascade: three positive measurements, each overturned

We arrived at the controlled result through a sequence of measurements of increasing rigor, each of which caught a confound in the previous one (Table 3). We present the cascade in full because it is itself a contribution: it shows concretely how intuitive measurements of representational drift produce false positives, and which control removes each.

Step 1 (under-power). A consecutive-turn drift with only 32 probes at a single layer is inconclusive: the noise floor at 32 probes (0.31, Table 1) is comparable to any motion present, so nothing can be concluded. This is why power must be fixed first.

Step 2 (no-context onset, a false positive). Measuring cumulative drift from turn 0 produces a large, apparently positive signal. The tell is that the between-dialogue spread (≈ 0.37) is far below the turn-0 displacement (≈ 0.88): almost all of the motion is the generic bare-probe→probe-in-context jump, which is shared across dialogues and is not content-conditional. Baselining at turn 1 and measuring $d(U_1, U_t)$ removes it.

Step 3 (same-context floor, a false positive over 21 layers). Comparing the between-dialogue spread against a same-context resampling floor looks positive at 21 layers, but this comparison is near-tautological. The floor resamples probes within one fixed context and therefore contains zero between-dialogue length, topic, and position variance, while the spread contains all of it. Empirically the between-dialogue spread grew monotonically with turn index in lockstep with the per-turn token-length gap, which widened from 3 to 20 tokens across turns ($\approx 12\%$ at the turn where the spread is measured): the signature of a length artifact, not of content.

Step 4 (small-sample placebo, a false positive at layers 11–21). Introducing the scrambled-twin null with $n=3$ dialogues appears to rescue a content effect at the mid-to-late layers (layers 11–21, real > scrambled; Figure 2). But this is a point estimate with no confidence interval, and its median rests on a single topically-divergent dialogue. Resampling at the dialogue level dissolves it.

Step 5 (the controlled result). With $n=12$ dialogues, the scrambled-twin null, and a dialogue-level bootstrap CI, the verdict is negative. This is the controlled measurement we report next.

Layer	Real spread [95% CI]	Scrambled spread [95% CI]	Clears rule?
12	0.347 [0.27, 0.47]	0.322 [0.29, 0.37]	no
15	0.323 [0.24, 0.41]	0.272 [0.24, 0.32]	no

Table 4: Controlled result on **Qwen2.5-0.5B-Instruct** (raw activations, 24 layers, $n=12$ dialogues, rank-1, dialogue-level bootstrap). Representative mid-late layers: real and scrambled-twin CIs overlap, no layer clears the decision rule, and no ≥ 3 -contiguous band forms. Verdict: NOT_CONTENT_CONDITIONAL.

Layer	Real spread [95% CI]	Scrambled spread [95% CI]	Clears rule?
1	0.653 [0.56, 0.69]	0.665 [0.57, 0.72]	no
late layers	real slightly > scrambled	CIs overlap throughout	no

Table 5: Cross-scale replication on **Qwen2.5-1.5B-Instruct** (raw activations, 28 layers, $n=12$, rank-1). The content-conditional band is empty; verdict NOT_CONTENT_CONDITIONAL.

5.4 Controlled result and replications

Model and data. We report **Qwen2.5-0.5B-Instruct** (Qwen Team, 2024) (896-dim residual stream, 24 layers), rank-1 difference-of-means directions, 64–96 probe pairs per side, and 12 topically diverse 5-turn dialogues. For the headline test we take each dialogue’s final-turn (full-context) hallucination subspace H_t at every layer, compute the median pairwise between-dialogue chordal spread, and bracket it with a dialogue-level bootstrap 95% CI, against the length- and topic-matched scrambled-twin null.

The hallucination subspace is not content-conditional on this model. The apparent content-conditionality does not survive (Figure 1, Table 4). At *every* layer the real-dialogue spread CI overlaps the scrambled-twin spread CI: at layer 12, real 0.347 [0.27, 0.47] versus scrambled 0.322 [0.29, 0.37]; at layer 15, real 0.323 [0.24, 0.41] versus scrambled 0.272 [0.24, 0.32]. A small, consistent, non-significant trend remains at the mid-to-late layers (real slightly above scrambled), but no layer satisfies the decision rule (real CI lower bound > scrambled CI upper bound), so no contiguous band exists and the verdict is NOT_CONTENT_CONDITIONAL. Scrambled twins, which hold token length, position, and lexical topic fixed while destroying coherent dialogue meaning, move the hallucination subspace as much as the real dialogues do. The between-dialogue movement is therefore attributable to length and topic, not to coherent dialogue content.

The safety side is worse, not better. The premise that matters for conditional *disentanglement* concerns the joint geometry of S_t and H_t . On this model the refusal direction is not cleanly recoverable below the mid layers: two independent estimates of the same S subspace are near-orthogonal (floor ≈ 1.0) through roughly the first half of the network, becoming recoverable only at the later layers. The cross-subspace overlap $\Omega(S_t, H_t)$ is therefore meaningful only at late layers, where its movement across turns is small. We thus cannot even measure the entanglement drift that conditional disentanglement is designed to track, let alone show it is large.

Replication across model scale (Qwen2.5-1.5B). The same controlled test on **Qwen2.5-1.5B-Instruct** (28 layers, $n=12$, rank-1) returns the same verdict: the content-conditional band is empty (Table 5). At layer 1, real 0.653 [0.56, 0.69] versus scrambled 0.665 [0.57, 0.72]; at the late layers real slightly exceeds scrambled but the CIs overlap throughout, so no layer clears the rule. The null is therefore not an artifact of the smallest model.

Replication in SAE feature space (GPT-2 small). Because disentanglement methods operate on SAE features rather than raw activations, we repeated the controlled test in the feature space of GPT-2 small using the publicly released residual-stream SAEs at all twelve layers (Cunningham et al., 2023; Bricken et al., 2023), encoding each contrastive probe’s last-token activation through the frozen SAE and recovering H_t in feature coordinates. The verdict is unchanged: at no layer does the real between-dialogue spread clear the

Layer	Real spread [95% CI]	Scrambled spread [95% CI]	Scrambled > real?
10	0.515 [0.46, 0.58]	0.659 [0.58, 0.80]	yes
11	0.541 [0.47, 0.64]	0.668 [0.59, 0.74]	yes

Table 6: SAE-feature-space replication on GPT-2 small (`gpt2-small-res-jb` residual SAEs, all 12 layers, $n=12$, rank-1). The band is empty, and at the late layers the scrambled-twin spread *exceeds* the real spread: length and lexical content explain the movement at least as well as coherent content. Verdict NOT_CONTENT_CONDITIONAL.

Layer	Hallucination subspace		Safety subspace	
	Real [95% CI]	Scrambled [95% CI]	Real [95% CI]	Scrambled [95% CI]
3	0.812 [0.73, 0.86]	0.790 [0.74, 0.87]	0.612 [0.53, 0.87]	0.587 [0.47, 0.75]
6	0.765 [0.70, 0.85]	0.870 [0.84, 0.93]	0.961 [0.92, 1.00]	0.922 [0.76, 1.00]
9	0.787 [0.71, 0.84]	0.855 [0.82, 0.91]	0.616 [0.49, 0.68]	0.677 [0.58, 0.87]
12	0.852 [0.78, 0.92]	0.888 [0.82, 0.92]	0.613 [0.55, 0.72]	0.539 [0.46, 0.68]
15	0.561 [0.50, 0.72]	0.484 [0.43, 0.58]	0.512 [0.46, 0.61]	0.433 [0.37, 0.50]
18	0.543 [0.41, 0.61]	0.416 [0.35, 0.48]	0.437 [0.38, 0.50]	0.390 [0.32, 0.46]
21	0.481 [0.37, 0.54]	0.325 [0.27, 0.42]	0.402 [0.36, 0.50]	0.336 [0.27, 0.40]

Table 7: Instruction-tuned SAE-feature-space replication on Gemma-2-2B-it (Gemma Scope width-16k residual SAEs, $n=12$, rank-1). Both the hallucination and the safety/refusal subspace return NOT_CONTENT_CONDITIONAL: no layer has the real spread’s CI lower bound above the scrambled spread’s CI upper bound. This is the only setting in which the safety subspace is recoverable enough to test, and it too shows no content-conditionality.

scrambled-twin spread, the band is empty, and at the late layers (7–11), where abstract features concentrate, the scrambled spread is in fact *larger* than the real spread (Table 6). At layer 10, real 0.515 [0.46, 0.58] versus scrambled 0.659 [0.58, 0.80]; at layer 11, real 0.541 [0.47, 0.64] versus scrambled 0.668 [0.59, 0.74]. The apparent context-dependence is, if anything, *better* explained by length and lexical content in feature space than in raw space. The null is thus not an artifact of the activation basis: it holds in the exact representation conditional disentanglement would use.

Replication on an instruction-tuned model, both subspaces (Gemma-2-2B-it). The settings above leave one gap: the safety/refusal subspace, poorly recoverable in the small raw models, and an instruction-tuned model in SAE feature space. We close both with Gemma-2-2B-it and the Gemma Scope residual SAEs (width 16k) at seven layers, testing the hallucination *and* the safety subspace (Table 7). Both return NOT_CONTENT_CONDITIONAL: no layer clears the rule for either subspace, and as in GPT-2 the scrambled twin matches or exceeds the real spread at several mid layers (hallucination layer 6: real 0.765 [0.70, 0.85] versus scrambled 0.870 [0.84, 0.93]; layer 9: 0.787 versus 0.855). The 16,384-dimensional SAE feature space has a higher estimation-noise floor (wider CIs, spreads of 0.4–0.9), so the test is less powered here; the finding is the absence of any real-over-scrambled gap, not a tight bound. Caveat: the public Gemma Scope SAEs are trained on the base model, so applying them to the instruction-tuned model is a mild distribution mismatch and these results are indicative.

5.5 Demonstration: the static baseline does not genuinely fail

The premise’s operational consequence is that a *static* disentangling projector, fit once, should fail at later turns, which is what motivates replacing it with a conditional one. We test this directly on `Qwen2.5-0.5B-Instruct` ($n=12$). The static projector built at turn 1 nulls the hallucination subspace H_1 ; its residual re-entanglement leak at turn t , the fraction of H_t the frozen projector fails to remove, is

Layer	Real leak $d(H_1, H_t)$ [95% CI]	Scrambled leak [95% CI]	Real > scrambled?
8	0.423 [0.41, 0.43]	0.464 [0.44, 0.48]	no
12	0.437	0.449	no

Only 3/25 layers (late: 18, 19, 21) cross the dialogue-level bar; *not* contiguous (≥ 3 rule unmet; chance ≈ 1.25).

Table 8: Static-projector demonstration on **Qwen2.5-0.5B-Instruct** ($n=12$). The static projector’s apparent across-turn re-entanglement is matched by its length/topic scrambled twin, so it does not genuinely fail. Verdict `STATIC_FAILURE_IS_CONFOUND`.

exactly $d(H_1, H_t) \in [0, 1]$, while a conditional projector rebuilt at turn t has leak 0 by construction. Naively the static projector looks broken: its leak by turn 4 is large, 0.3–0.6 across layers (Figure 3). But the length- and topic-matched scrambled twins leak *just as much* (Table 8): at layer 8, real 0.423 [0.41, 0.43] versus scrambled 0.464 [0.44, 0.48]; at layer 12, real 0.437 versus scrambled 0.449. Only 3 of 25 layers (all late: 18, 19, 21) show real leak exceeding scrambled at the dialogue-level 95% level, and they are not contiguous, below our pre-registered ≥ 3 -contiguous-layer rule and near the ≈ 1.25 layers expected by chance. We report this weak, non-robust late-layer elevation rather than hide it. The verdict is `STATIC_FAILURE_IS_CONFOUND`: for this behavior and model, a static projector’s degradation across turns is confound-explained, so a content-conditional projector would not fix it and the conditional machinery addresses a confound rather than a real non-stationarity.

Summary across settings. Every configuration tested under the full protocol returns `NOT_CONTENT_CONDITIONAL`: **Qwen2.5-0.5B-Instruct** and **Qwen2.5-1.5B-Instruct** in raw activation space, GPT-2 small in SAE feature space, and **Gemma-2-2B-it** in SAE feature space for *both* the hallucination and the safety/refusal subspace; the static-projector demonstration returns `STATIC_FAILURE_IS_CONFOUND`. Across these models, in raw and SAE feature space, and for both subspaces, the apparent context-conditionality is statistically indistinguishable from a length and topic artifact.

6 Discussion, Scope, and Limitations

What this does and does not establish. It establishes that, for the model and construction tested, three intuitive measurements of context-conditionality are confounded, and that the controlled measurement returns a null. It does *not* establish that the premise is universally false. We have closed the most important cells and mark the remainder. The instruction-tuned SAE-feature-space cell is now run on Gemma-2-2B-it for *both* the hallucination and the safety/refusal subspace (Section 5, Table 7), the only setting in which the safety subspace is recoverable enough to test, and it too returns the null; and the static-projector mis-specification test (Section 5.5) shows the static baseline does not genuinely fail. The genuinely open cells are: **(a)** larger instruction-tuned models and a mixture-of-experts model (for the per-expert variant of the premise); **(b)** higher-rank subspaces (rank-1 is the powered default); **(c)** a within-topic content manipulation that separates epistemic state from surface topic; and **(d)** a behavioral, generation-time static-projector test. A caveat on the Gemma cell: the public Gemma Scope SAEs are base-model-trained, so the instruction-tuned run is a mild distribution mismatch and is indicative. Our protocol ports to all of these with only a change of backend.

Residual confounds in the control. The scrambled twin holds token length and topic but conflates coherent “dialogue content” with “epistemic state”; a fully within-topic content manipulation would separate these. Word-shuffling may also partially degrade topic extraction; this would, if anything, bias toward a (spurious) positive, which we do not observe.

Implication. The negative result is not an indictment of conditional steering or disentanglement; it is a request that such methods measure, with controls, the premise they assume. A static intervention is mis-

specified only if the geometry it freezes actually moves with context beyond length and topic. On the present evidence, the cheap and decisive thing is to run the controlled measurement first.

7 Lessons and Actionable Recommendations

The point of this study is not the single null but the transferable lessons that produced it; we state them as recommendations for two audiences.

For anyone measuring whether a manipulation changes a representation (context, in-context examples, a steering edit, a fine-tune), the following four controls flip conclusions in practice and should be standard:

1. **Use a matched-null, not a same-condition resampling floor.** A bootstrap over probes at a fixed condition contains none of the between-condition variance (length, position, topic) that the test statistic carries, so “effect > floor” is near-tautological. Build a null that matches the nuisance variables and differs only in the variable of interest (here, length/topic-matched scrambled twins). This single change converted our headline from positive to null.
2. **Measure with a gauge-invariant quantity.** Subspaces are recovered up to an arbitrary basis; a basis-dependent “drift” conflates estimator gauge with real movement. Use principal-angle/chordal distances.
3. **Bootstrap over the unit of generalization, not the convenient one.** A confidence interval over probes says nothing about generalization across dialogues; resample the dialogues. Our mid-layer “effect” was significant over probes and vanished over dialogues ($n=3 \rightarrow n=12$).
4. **Baseline away trivial transitions.** The no-context $\rightarrow$ context jump is large, shared, and uninteresting; measure drift relative to the first in-context turn, and report the onset separately.

For designers of conditional alignment methods (conditional steering, conditional disentanglement): a conditional intervention is justified only if the geometry it conditions on *moves* with context beyond nuisance confounds, and if a static baseline *fails*. We recommend reporting, as a precondition, (i) the controlled context-conditionality measurement above, in the feature space the method uses, and (ii) a static-projector baseline tested for content-driven re-entanglement. We carried out exactly this test (§5, Figure 3): the static projector’s across-turn leak is fully matched by the length/topic placebo, so it does not genuinely fail. On the present evidence the conditional machinery, for the behaviors and models studied, addresses a confound rather than a real non-stationarity. We do not claim this generalizes to all settings; we claim that without this controlled check a method cannot tell the two apart, and the check is cheap.

A reusable checklist. The released harness packages the above as a drop-in test: a gauge-invariant drift metric, a scrambled-twin null, dialogue-level bootstrap CIs, a power curve, and a synthetic ground-truth validation, with a single backend swap between raw activations and SAE features. The intended use is pre-registration: run it, report the controlled measurement and the power, and only then build (or not).

8 Conclusion

We asked whether dialogue context rotates the safety and hallucination subspaces of an LLM, the premise underlying conditional disentanglement. The obvious measurements say yes; each is a confound. Under a length/topic-matched, dialogue-level-bootstrapped protocol the effect is within noise on the model tested, and the safety subspace is not recoverable where the premise would need to hold. We release the protocol and the full rigor cascade so that representational-drift claims, positive or negative, can be vetted before they are built upon.

Reproducibility. All code, probe banks, dialogues, cached activations, and the synthetic ground-truth validation are released; every figure and number is regenerated by the released scripts.

Cell	Objection it closes	Status
Qwen2.5-0.5B-Instruct, raw acts, rank-1	baseline	done
Qwen2.5-1.5B-Instruct, raw acts, rank-1	“only the smallest model”	done
GPT-2 small, SAE feature space, rank-1	“raw acts aren’t where methods operate”	done
static-projector demonstration (Qwen2.5-0.5B)	“you never tested the actual claim”	done
Gemma-2-2B-it, SAE space, hallucination + safety	“SAE safety side on a chat model”	done
mixture-of-experts model (per-expert premise)	per-expert variant of the premise	pending
rank-2/3 subspaces	“rank-1 is too crude”	pending (needs hundreds of probes)
within-topic content control	“topic conflated with content”	pending
behavioral (generation-time) static-projector test	generation-time mis-specification	pending (future work)

Table 9: Evaluation matrix. Raw-activation replication across two model scales, the SAE feature-space replication (GPT-2 and Gemma-2-2B-it, the latter for both the hallucination and the safety subspace), and the static-projector demonstration are complete. The remaining cells (mixture-of-experts, higher rank, within-topic control, behavioral test) are pending; the protocol ports to each with only a backend swap.

A Synthetic Validation Details

The synthetic test (Table 2) runs with no model and no network and reproduces on any machine, so it isolates the estimator and decision rule from any property of a particular language model. We construct paired activations whose contrast subspace is an analytically known object, then check four properties.

Construction. For each turn we draw class-conditional activation clouds whose difference-of-means defines a rank- k ground-truth subspace with a prescribed orientation. Two ground-truth trajectories are used: a *static* subspace whose orientation is fixed across turns, and a *rotating* subspace whose orientation advances by a fixed 15° per turn. Estimation noise is controlled by the number of probe pairs, matching the power regime used on real data (§5.2).

What it confirms. (i) At high SNR the estimator recovers the injected subspace. (ii) The static trajectory returns measured drift at the bootstrap noise floor, and the decision rule returns negative, the property that prevents a low-power run from being read as a positive. (iii) The 15° /turn rotation is recovered as noise-corrected chordal drift 0.144 against the analytic 0.149, clears the floor, and the rule returns positive; a cross-subspace overlap with a known time-varying relative angle is tracked over the same trajectory. (iv) Principal-angle sanity: identical planes return 0, orthogonal planes return $\pi/2$. Because the identical code path is used on real data, a verdict on a real model reflects the data rather than a code artifact.

B Evaluation Matrix: Done and Pending

Table 9 records the evaluation matrix. The acceptance risk for a scoped negative is not novelty but breadth, the objection that the effect was missed because of the wrong model, space, or rank; we therefore enumerate the cells that close that objection and mark each as done or pending, with the released harness producing the statistic for both. The single most important such cell, the safety subspace in SAE feature space on an instruction-tuned chat model, is now complete: on Gemma-2-2B-it with Gemma Scope (Lieberum et al., 2024; run in bfloat16 on an L4/A100, after a `transformer_lens` plus Gemma-2 load issue ruled out the T4), both the hallucination and the safety subspace return the null (Table 7). The remaining pending cells are a mixture-of-experts model, higher-rank subspaces, a within-topic content control, and a behavioral static-projector test.

C Probe-Bank Construction

Contrastive pairs. Subspaces are recovered by contrast, following standard practice in the probing and steering literature. The *hallucination/epistemic* subspace H_t is the rank-1 difference-of-means direction between answerable real-entity questions and surface-matched unanswerable fictional-entity questions, following

the knowledge-awareness construction of Ferrando et al. (2025). The *safety* subspace S_t is the analogous contrast between harmful requests and surface-matched harmless requests, the refusal direction of Arditi et al. (2024). Pairs are surface-matched so that the recovered direction isolates the epistemic or safety contrast rather than lexical or syntactic differences.

Fixed probes, varying context. At dialogue turn t we prepend the running conversation (the first t user turns, each followed by a fixed neutral assistant reply) and append the contrastive probe bank as the next user turn, reading the last-token residual activation at each layer. The probes are held fixed and only the prepended context varies, so any movement in the recovered U_t is attributable to context.

Power-driven sizing. The power analysis (§5.2) requires ≥ 96 probe pairs per side at rank-1; we use 40–96 pairs per side in the reported runs. The deliberately small banks used for laptop development are not large enough for a publishable verdict, which is precisely why the floor is reported with every drift.

Scrambled-twin null. The length- and topic-matched null is generated by word-shuffling each user turn. This preserves the exact token multiset (and hence length, topic, and lexicon) while destroying coherent meaning, so genuine content-conditionality requires the real subspace to move more than its scrambled twin. As noted in the scope discussion, the scrambled twin holds topic fixed but conflates coherent dialogue content with epistemic state; a fully within-topic manipulation (a pending cell) would separate the two.

D Reproducibility Statement

All code, probe banks, dialogue scripts, cached activations, and the synthetic ground-truth validation are released in a public repository. Every figure and number in this paper, including the synthetic validation (Table 2), the power curve (Table 1), the rigor cascade (Table 3), the controlled result and its cross-model and SAE-feature-space replications (Tables 4–6), the static-projector demonstration (Table 8), and Figures 1, 2, and 3, is regenerated by the released scripts.

Pipeline. The estimator and decision rule are validated with a model-free synthetic test that reproduces anywhere. Activation extraction reads the last-token residual stream at each layer; the SAE-feature-space backend passes that activation through the frozen released SAE encoder and runs the identical contrast-and-drift pipeline on the feature vector, so the only change between the raw and SAE settings is a backend swap. Drift is computed as the gauge-invariant chordal distance with dialogue-level bootstrap confidence intervals, and the pre-registered decision rule (real CI lower bound $>$ scrambled CI upper bound over a ≥ 3 -contiguous-layer band) is applied uniformly across all settings.

Determinism and scope. Random seeds for the bootstrap and for synthetic data generation are fixed, so the reported intervals reproduce. The released harness packages the gauge-invariant drift metric, the scrambled-twin null, dialogue-level bootstrap CIs, the power curve, and the synthetic validation as a drop-in test with a single backend swap between raw activations and SAE features, so that a representational-drift claim, positive or negative, can be vetted before it is built upon. All results are scoped to the models and feature spaces evaluated here, `Qwen2.5-0.5B-Instruct` and `Qwen2.5-1.5B-Instruct` in raw activation space and GPT-2 small in SAE feature space; we do not claim the verdict generalizes beyond them.

References

- Andy Arditi, Oscar Obeso, Aaqib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.
- Alan Edelman, Tomás A Arias, and Steven T Smith. The geometry of algorithms with orthogonality constraints. *SIAM Journal on Matrix Analysis and Applications*, 20(2):303–353, 1998.
- Javier Ferrando, Gabriele Sarti, Arianna Bisazza, and Neel Nanda. Do I know this entity? knowledge awareness and hallucinations in language models. In *International Conference on Learning Representations (ICLR)*, 2025.
- Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, et al. Scaling and evaluating sparse autoencoders. *arXiv preprint arXiv:2406.04093*, 2024.
- Subhash Kantamneni, Joshua Engels, Senthooan Rajamanoharan, Max Tegmark, and Neel Nanda. Are sparse autoencoders useful? a case study in sparse probing. In *International Conference on Machine Learning (ICML)*, 2025.
- Adam Karvonen, Can Rager, Johnny Lin, Curt Tigges, Joseph Bloom, et al. SAE Bench: A comprehensive benchmark for sparse autoencoders. In *International Conference on Machine Learning (ICML)*, 2025.
- Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Tom Lieberum, Senthooan Rajamanoharan, Arthur Conmy, Lewis Smith, et al. Gemma scope: Open sparse autoencoders everywhere all at once on Gemma 2. In *BlackboxNLP Workshop (EMNLP)*, 2024.
- Stephanie Lin, Jacob Hilton, and Owain Evans. TruthfulQA: Measuring how models mimic human falsehoods. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022.
- Omar Mahmoud, Ali Khalil, Buddhika Laknath Semage, Thommen George Karimpanal, and Santu Rana. The unintended trade-off of AI alignment: Balancing hallucination mitigation and safety in LLMs. *arXiv preprint arXiv:2510.07775*, 2025.
- Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in LLM representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2023.
- Wenbo Pan, Zhichao Liu, Qiguang Chen, Xiangyang Zhou, Haining Yu, and Xiaohua Jia. The hidden dimensions of LLM alignment: A multi-dimensional analysis of orthogonal safety directions. In *International Conference on Machine Learning (ICML)*, 2025.
- Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- Senthooan Rajamanoharan, Tom Lieberum, Nicolas Sonnerat, Arthur Conmy, et al. Jumping ahead: Improving reconstruction fidelity with JumpReLU sparse autoencoders. *arXiv preprint arXiv:2407.14435*, 2024.
- Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.
- Paul Röttger, Hannah Rose Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In *NAACL*, 2024.

- Daniel Tan, David Chanin, et al. Analysing the generalisation and reliability of steering vectors. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, et al. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. *Transformer Circuits Thread*, 2024.
- Tom Wollschläger, Jannes Elstner, Simon Geisler, Vincent Cohen-Addad, Stephan Günnemann, and Johannes Gasteiger. The geometry of refusal in large language models: Concept cones and representational independence. In *International Conference on Machine Learning (ICML)*, 2025.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.